



this is my docker-compose.yml "services:

mysql-db:

image: mysql:8.0

env_file:

- .env

ports:

- "3307:3306"

volumes:

- mysql_data:/var/lib/mysql

backend-subscriber:

build: ./backend

container_name: backend-subscriber

env_file:

- .env

depends_on:

- mysql-db

ports:

- "5000:5000"

react-dashboard:

build: .

container_name: react-dashboard

ports:

- "3000:80"

depends_on:

- backend-subscriber

volumes:

mysql_data:", this is my dockerfile in root folder "# Stage 1: build React app

FROM node:16-alpine AS build

WORKDIR /app

COPY package.json package-lock.json ./

RUN npm install

COPY . .

RUN npm run build

Stage 2: serve React app with nginx and SPA routing

```
FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"] And this is my .env
"MYSQL_ROOT_PASSWORD=Mycloud@25
MYSQL_DATABASE=vehicle_dashboard
MYSQL_USER=vehicledbuser
MYSQL_PASSWORD=Mycloud@25
```

```
DB_HOST=mysql-db
DB_USER=vehicledbuser
DB_PASSWORD=Mycloud@25
DB_NAME=vehicle_dashboard
```

```
REACT_APP_BACKEND_URL=http://localhost:5000"
```

Here is a concise and structured document summarizing your steps and the key configuration files for dockerizing your React + Flask + MySQL project based on the provided `docker-compose.yml`, `Dockerfile`, and `.env`:

Dockerizing a Full Stack React + Flask + MySQL Application

Overview

This document outlines the process and configuration used to containerize a multi-service application consisting of:

- A MySQL database
- A Python Flask backend API (backend-subscriber)
- A React frontend dashboard (react-dashboard)

Docker Compose is used to orchestrate container creation, networking, and environment configuration.

Key Configuration Files

docker-compose.yml

```
services:
  mysql-db:
    image: mysql:8.0
    env_file:
      - .env
    ports:
      - "3307:3306"
    volumes:
      - mysql_data:/var/lib/mysql

  backend-subscriber:
    build: ./backend
    container_name: backend-subscriber
    env_file:
      - .env
    depends_on:
      - mysql-db
    ports:
      - "5000:5000"

  react-dashboard:
    build: .
    container_name: react-dashboard
    ports:
      - "3000:80"
    depends_on:
      - backend-subscriber

volumes:
  mysql_data:
```

React Dockerfile (root folder)

```
# Stage 1: build React app
FROM node:16-alpine AS build
WORKDIR /app
COPY package.json package-lock.json ./
RUN npm install
COPY . .
RUN npm run build

# Stage 2: serve React app with nginx and SPA routing
FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

`.env`

```
MYSQL_ROOT_PASSWORD=Mycloud@25
MYSQL_DATABASE=vehicle_dashboard
MYSQL_USER=vehicledbuser
MYSQL_PASSWORD=Mycloud@25

DB_HOST=mysql-db
DB_USER=vehicledbuser
DB_PASSWORD=Mycloud@25
DB_NAME=vehicle_dashboard

REACT_APP_BACKEND_URL=http://localhost:5000
```

Important Notes & Steps To Run

- Set `REACT_APP_BACKEND_URL` to your Flask backend's accessible address from the browser (usually `http://localhost:5000` for local development).
- Build all images with: `docker-compose build`
- Run containers in detached mode: `docker-compose up -d`
- Confirm containers are running: `docker ps`
- React frontend is available at `http://localhost:3000`
- Flask backend API is listening at `http://localhost:5000`
- MySQL data persists to a Docker named volume `mysql_data`

Troubleshooting

- React loading indefinitely usually indicates React cannot reach backend API.
- Confirm API URL in `.env` and rebuild React image after any `.env` changes.
- Use browser dev tools console and network tabs to debug fetch requests and errors.
- Ensure backend Flask API serves data correctly (`/latest` endpoint).
- Use `docker logs` to diagnose individual container issues.

This document can serve as your project README or guide for teammates to understand or setup the Dockerized environment you created.

If you want, this can be further expanded with CI/CD, deployment tips, or advanced Docker Compose options.

Let me know if you want help creating a nicely formatted PDF or Markdown file from this content!

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2. <https://roytuts.com/docker-compose-dockerizing-flask-mysql-app/>
3. <https://stavshamir.github.io/python/dockerizing-a-flask-mysql-app-with-docker-compose/>
4. <https://stackoverflow.com/questions/53235312/docker-compose-getting-mysql-db-fully-started-before-flask-app-starts>
5. https://www.linkedin.com/posts/renuka-vaishnav-715880249_docker-flask-mysql-activity-7341140206701432834-1YUL
6. https://www.reddit.com/r/flask/comments/10ihdza/how_to_deploy_a_flask_react_mysql_app/
7. <https://www.squash.io/tutorial-building-a-chat-web-app-using-flask-mongodb-reactjs-bootstrap-and-docker-compose/>